

Are simple technical trading rules profitable in bitcoin markets?

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Abstract

This paper examines the profitability of simple technical trading rules in bitcoin markets comprehensively, by taking into account realistic investor behavior and transaction costs, and data mining problems. Realistic investor behavior is replicated by first employing 75,360 simple technical trading rules, divided over 6 commonly used trading rule classes and daily and intraday frequencies. Next, we select the best performing rules after transaction costs using a multiple hypothesis procedure. Finally, we form portfolios combining the selected rules and analyse their out-of-sample performance. We find that, especially risk-return wise, simple technical trading rules can outperform a buy-and-hold strategy in the bitcoin market out-of-sample.

Keywords: Bitcoin, Technical Analysis, False discovery rate, Intraday

JEL: G11, G14, G17

1. Introduction

In 2008 in the midst of the global financial crisis and amidst growing public distrust in financial institutions, Satoshi Nakamoto (2008) introduced a peer-to-peer electronic cash system, named Bitcoin. Two key characteristics of Bitcoin are that it does not need any trusted third party in order to process electronic payments, and that it has a supply regulated by a fixed, known algorithm, attempting to reduce the influence of financial intermediaries and central authorities. More than a decade since its inception, Bitcoin is well known by the general public, mainly due to its exponential increase in value. However, as a relatively new asset without any apparent intrinsic value, the bitcoin market has been plagued by high volatility, market inefficiencies and bubbles (Cheah and Fry, 2015; Urquhart, 2016; Bariviera et al., 2017; Nadarajah and Chu, 2017; Kristoufek, 2018; Sensoy, 2019). Following the efficient market hypothesis, these circumstances in the market should theoretically make bitcoin returns more predictable and the use of technical analysis profitable.

At the time of writing some research has already been done regarding profitability of technical trading rules in bitcoin markets. A short overview of the existing literature is given in table 1. First we take a look at the studies using daily price data. Gerritsen et al. (2020) examine the Sharpe ratios of 7 trend-following indicators between 2010 and 2018 using daily price data. They find that mainly support-and-resistance (S&R) rules outperform a buy-and-hold strategy in terms of risk-adjusted measures during their full sample period, mainly during bull or bear markets. Grobys et al. (2020) test a small set of moving average (MA) rules between 2016 and 2018 across 11 cryptocurrencies using a multivariate test. They find return predictability across different cryptocurrencies, irrespective of whether bitcoin is accounted for. Detzel et al. (2021) propose a new equilibrium theory that shows that price-to-MA ratios can forecast returns of assets with hard-to-value fundamentals. To confirm their model empirically, a study evaluating a limited sample of MA strategies between 2010 and 2018 is conducted. They find that simple MA rules outperform, with mainly a significant Sharpe ratio and/or alpha, a buy-and-hold strategy out-of-sample (OOS). The trading rules have a better performance before 2014, which is in line with the literature finding that the bitcoin market is becoming more

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Paper	Sample Period	Freq.	Prof.	OOS	TC	MHT	#TTR	TTR
Corbet et al. (2019)	01/14 - 06/18	1 min.	α	-	-	X	548	<u>MA</u> , S&R
Miller et al. (2019)	02/18 - 03/18	1 min.	α	-	-	-	120	chart pat- terns
Gerritsen et al. (2020)	07/10 - 12/18	Daily	SR	-	-	-	20	MA, <u>S&R</u> , RSI, MACD, ROC, OBV, BB
Grobys et al. (2020)	01/16 - 12/18	Daily	α	-	-	-	5	<u>MA</u>
Detzel et al. (2021)	07/10 - 06/18	Daily	α & SR	-	BE	-	5	<u>MA</u>
Hudson and Urquhart (2021)	07/10 - 12/17	Daily	SR	X	BE	X	14,919	MA, S&R, RSI, <u>Filter</u> , <u>CB</u>
Svogun and Bazán-Palomino (2022)	01/16 - 11/21	1 min, Daily	α & SR	-	X	-	69	<u>MA</u> , S&R

Table 1: Overview of the literature. This table summarizes the differences between the data and methodology used by papers analysing technical trading rules in bitcoin markets. We compare the sample period, the trading frequency analysed, if profitable trading rules were found (with α and SR meaning profitability in absolute returns or risk-adjusted measures, respectively), if trading rules were analysed out-of-sample, if transaction costs were taken into consideration (with BE indicating that break-even transaction costs were used), if multiple hypothesis tests (MHT) controlling for data mining were used, the amount of trading rules analysed and the classes of trading rules analysed (with the most profitable class underlined), respectively.

efficient over time. Hudson and Urquhart (2021) test 14,919 technical trading rules using multiple hypothesis procedures between 2010 and 2017. They compare 5 classes of trading rules and find that all classes improve the risk-adjusted performance in cryptocurrency markets, with filter and channel breakout (CB) rules having the best performance. After calculating break-even transaction costs, they conclude that their results for S&R rules would not hold in realistic market conditions. However, they do not find out-of-sample (OOS) outperformance for bitcoin. As far as we know, three studies assess the profitability using high-frequency trading data. Corbet et al. (2019) test a small set of MA rules and S&R rules using different intraday price data between 2014 and 2018. They find that the variable MA strategy is most profitable on the 1-minute trading frequency. Miller et al. (2019) use 1-minute high frequency data between February and March 2018 to identify various possible chart patterns (e.g. head-and-shoulders pattern). They find that some patterns result in great absolute returns over a buy-and-hold benchmark. Svogun and Bazán-Palomino (2022) test 69 MA and S&R rules on the 1-minute and daily trading frequencies for 5 different cryptocurrencies between 2016 and November 2021. They look at performance after transaction costs, which are approximated by fees paid to miners ¹. They find that technical trading rules can outperform a buy-and-hold benchmark in the bitcoin market, and furthermore that trading on a daily basis is more robust to transaction costs. For bitcoin, trading rules are more likely to outperform the benchmark during downturns and not during bubbles.

Generally these studies show that there is evidence of profitability of technical trading rules for different trading frequencies and technical trading rule classes in the bitcoin market. However, looking at table 1 we see some possibilities to extend the existing literature. First, only one study has a sample period that ends after 2018. Bitcoin, being a relatively new asset, has undergone various changes, most interestingly in terms of market efficiency and conditions, availability of derivatives and regulations, throughout the years. The popularity of bitcoin soared at the end of 2017², and although the general public interest reached its peak in 2017, it stayed at a more elevated level than before 2017. Various studies find that the bitcoin market is becoming more efficient over time (Nadarajah and Chu, 2017; Bariviera et al., 2017; Sensoy, 2019), and

¹These fees are representative for trading costs when you exclusively trade on the blockchain. Transactions on specialized online cryptocurrency exchanges do not incur mining fees, but rather traditional transaction fees and costs due to bid-ask spreads.

²As measured by Google Trends.

shorting, margin and derivative trading have been introduced on various exchanges after 2017, therefore an updated, longer sample period may be of interest. Second, research has shown that technical analysis is used by professionals in the foreign exchange market mainly on the short and medium term (Menkhoff and Taylor, 2007; Park and Irwin, 2007). Despite this, only 3 studies use intraday price data, but at a very high frequency (1-minute). Third, when evaluating the performance of the trading rules, not all studies take into account: (1) The risk-adjusted performance, which may be of interest for portfolio management (2) Transaction costs, which severely limit the profitability of trading rules with many transactions. Typically there is a trade-off between more closely following the price due to more trading signals and higher transaction costs. A recent study shows that omitting costs from your analysis can inflate false discoveries (Anghel, 2022) (3) OOS performance and multiple hypothesis tests, which help in limiting spurious results from data mining. (4) some studies look at a limited set of trading rules, while there exists many popular classes with a limitless amount of parameterizations. Realistically, investors combine signals from these different trading rules and trading frequencies, usually selecting the best performing rules after transaction costs and after good OOS performance. We add to the existing literature by combining a wide range of technical trading rules across intraday and daily trading frequencies, mimicking more realistic investor behavior, and testing their profitability after transaction costs and OOS, while taking into account data mining.

2. Data

We use tick-by-tick transaction data from Bitstamp³, a online cryptocurrency exchange, which contains the volume and exchange rate of each trade of the BTC-USD pair between 2012 and 2022. After cleaning the data with the method described in Brownlees and Gallo (2006)⁴, we re-sample it at a daily, 60, 30 and 10 minute frequency, with the daily exchange rate being determined at midnight UTC. We use the period before 2013 to initialize our technical trading rules and the rest of the sample for our analysis. We convert the prices, P_t , in our sample to logarithmic returns such that:

$$r_t = \ln \frac{P_t}{P_{t-1}} \quad (1)$$

Next to transaction data we also approximate 'realistic' transaction costs by using the average daily logarithmic bid-ask spread⁵, and the minimum transaction fee⁶ on the Bitstamp exchange.

Figure 1 shows the average, minimum, and maximum monthly BTC-USD exchange rate between 2012 and 2022. During our sample bitcoin has known an astronomical increase from 10 USD to 65,000 USD. This increase, however, is paired with volatile price swings, most notably at the end of 2013, 2017, and at the end of our sample. Table 2 shows the summary statistics of the returns for the 4 sampling frequencies and the 0- and 100-BTC-deep bid-ask spreads⁷. Looking at the returns, we see that the standard deviation is multiples of the mean return, showing the extreme volatility in the BTC market. The returns are negatively skewed on lower and positively skewed on higher trading frequencies. They are, as in previous studies, leptokurtic, but the kurtosis declines with temporal aggregation. The minimum and maximum returns are quite extreme for every sampling frequency, with the lower frequencies having lower minima, and the higher frequencies having higher maxima, which may explain the differences in skewness. There is a more logical difference between the first and third quartiles of the returns, but these are still quite high on the higher trading frequencies, showing that big price movements happen during very short time frames in the bitcoin

³The data is obtained via [dataset] Bitcoincharts.com (2022)

⁴First we delete all trades with a 0 or negative price or volume. Then we substitute the price of simultaneous trades with the median price at that time. Lastly we run their data cleaning algorithm in order to detect outliers, with $\delta = 10\%$, $\gamma = 1$, $k = 40$.

⁵The data is obtained via [dataset] Bitcoinity.org (2022)

⁶Bitstamp charges a uniform trading fee of minimum 0,20% per trade before March 2nd 2015 and 0,10% afterwards. In 2019 they lowered their fees for exceptionally high volumes even further to 0%, but we don't take this decrease into account in our paper (Bitstamp, 2012, 2015b, 2019).

⁷The X -BTC-deep bid-ask spread is the bid-ask spread calculated with the limit prices of the orders with a cumulative liquidity of X Bitcoins on both sides of the order book.

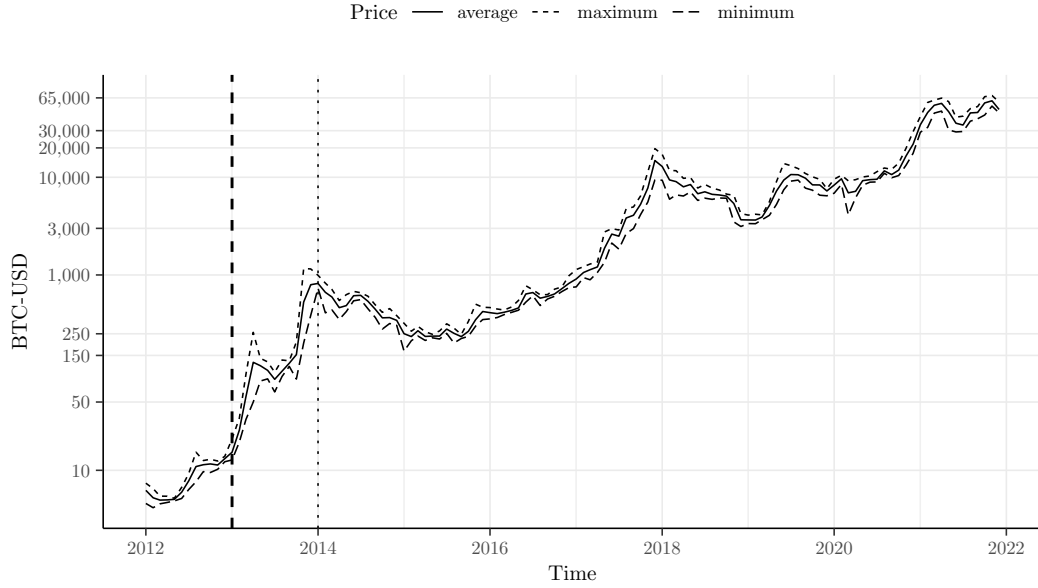


Figure 1: BTC-USD exchange rate on Bitstamp from 2012 until 2022. We report the monthly average, minimum, and maximum closing price. The dashed line indicates the start of our sample of technical trading rules. The dotted line indicates the start of our sample of OOS portfolios.

market. We also note that there exist some significant autocorrelation on different lags, which may point to return predictability in our sample. Interestingly, the autocorrelation is more positive on longer/higher and more negative on the lower lags and trading frequencies, hinting at possible short term trend reversal and longer term momentum effects in the returns. Looking at the bid-ask spreads, we see that the average 0-BTC-deep spread is 0.134%, while the average 100-BTC-deep spread is more than 1%-point higher. These spreads can be quite volatile as well⁸.

Using data from the Bitstamp exchange has 2 distinct advantages. First, Bitstamp is one of the oldest online trading platforms for cryptocurrencies. It was founded in 2011 and still exists today, resulting in one of the longest available datasets for bitcoin transaction data on a single platform. During its life, it had a stable market share of around 10% of the total exchange traded volume of BTC-USD ([dataset] Bitcoinity.org, 2022), representing a significant part of the market. Second, Bitstamp is associated with few incidents and trade disruptions⁹, resulting in a relatively continuous data set with little missing data and a good representation for the overall market price fluctuations. The platform does not provide margin trading and short-selling during our sample period, which will be taken into account in our study, and had a relatively stable fee system in place.

3. Methodology

In this paper we attempt to mimic more realistic investor behaviour and circumstances. Technical traders usually use various classes of technical trading rules and various trading horizons and frequencies concurrently. By combining the input from these two dimensions they decide whether to buy or sell a certain asset, giving more importance to rules with high historical outperformance (Menkhoff and Taylor, 2007). They aim to optimize their profits net of costs and OOS.

⁸The bid-ask spreads are much higher in the beginning of the sample, but they slowly decrease over time.

⁹In its long existence Bitstamp was only hacked once in 2015, when almost 19,000 Bitcoins were stolen (Bitstamp, 2015a)

	Day	60 Min.	30 Min.	10 Min.	0-spread	100-spread
N	3287	78888	157776	473328	3252	3252
Mean	0.248	0.01	0.005	0.002	0.134	1.495
Std. Dev.	4.61	1.053	0.798	0.5	0.124	1.316
Skewness	-1.368	-1.372	1.037	7.087	3.136	3.128
Kurtosis	23.171	107.405	408.931	1317.227	16.007	12.249
Min.	-66.395	-34.729	-44.287	-29.725	0.014	0.3
Q1	-1.367	-0.279	-0.209	-0.127	0.055	0.798
Median	0.208	0.009	0.003	0	0.095	1.054
Q3	2.024	0.315	0.229	0.138	0.169	1.564
Max.	33.749	29.888	59.83	73.691	1.505	12.886
$\rho(1)$	-0.023	-0.087***	-0.105***	-0.09***		
$\rho(2)$	0.001	-0.006*	-0.007***	-0.041***		
$\rho(5)$	0.068***	0.011***	-0.004	-0.006***		
$\rho(10)$	0.056***	0.007*	-0.003	-0.002		

Table 2: Descriptive statistics of the log return of the BTC/USD exchange rate per sampling frequency and the log bid-ask spread at the 0- and 100-BTC order book depth, on Bitstamp between 2013 and 2022. The returns and spreads are given in percentages (e.g. the average of 0.263 for the daily log return = 0.263%). We show the amount of observations (N), the arithmetic mean, standard deviation, skewness, kurtosis, minimum, 1st quartile, median, 3rd quartile and maximum. Lastly, $\rho(k)$ is the autocorrelation on the k th lag.

Keeping this in mind, we use 4 steps in order to determine if technical trading rules are profitable in bitcoin markets. (1) We determine the investment universe, a set of possible technical trading rules that an investor may consider. (2) For each sub-sample and for every technical trading rule we then calculate the daily return and appropriate performance measures (3) Using these performance measures, we determine which rules statistically outperform the benchmark during each sub-period (4) We form a portfolio from the outperforming technical trading rules and test if it is profitable OOS.

3.1. Technical trading rules

We construct 18,840 (75,360 over all 4 sampling frequencies) individual technical trading rules, subdivided over 6 commonly used classes of technical trading rules, namely: moving averages (MA), filter rules, support and resistance (S&R) trading rules¹⁰, channel breakout (CB) trading rules, on-balance volume averages (OBV), and the relative strength indicator (RSI). Our selection of technical trading rules is largely based on the selection of Sullivan et al. (1999) and Hsu et al. (2016). A more in depth discussion of the different parameterizations can be found in Appendix B.

3.2. Returns and performance measures

As mentioned in section 2, Bitstamp does not support short and margin trading. Therefore, we opt for an in-or-out trading strategy where we take a long position in bitcoin if a buy signal is given, and invest in the risk-free rate¹¹ otherwise. The return of the j -th technical trading rule is then calculated as follows:

$$r_{j,t} = D_{j,t-1}r_{b,t} + D_{j,t-1}^{rf}r_{rf,t} - C_{j,t} \quad (2)$$

with $D_{j,t-1} \in \{1, 0\}$ and $D_{j,t-1}^{rf} = \begin{cases} 1, & \text{if } D_{j,t-1} = 0 \\ 0, & \text{otherwise} \end{cases}$, for $j = 1, \dots, J$.

¹⁰Also called trading range breakout rules in other studies.

¹¹Similarly as in Gerritsen et al. (2020), the US 3-month treasury bill rate, obtained via [dataset] OECD (2022) and adjusted for trading frequency, is used as the risk-free rate. Although investing in the risk-free rate may not be economically, or practically feasible, especially at the higher trading frequencies, we still utilise them as they introduce some variation in the in-or-out strategies. Including the risk-free rate has a minimal impact on our results, considering the 3-month T-bills have a total return of 7.6% during our sample, compared to bitcoin's 365,288%.

Where $r_{b,t}$ and $r_{rf,t}$ are the logarithmic returns of the BTC-USD exchange rate and the risk-free rate respectively. $D_{j,t-1}$ and $D_{j,t-1}^{rf}$ are the 1-period position in BTC-USD or the risk-free asset of the j -th technical trading rule respectively. $C_{j,t}$ is the (logarithmic) trading cost at time t . It consists of a flat percentage fee and half of the bid-ask spread. After calculating the returns for every rule at every sampling frequency, all returns are re-sampled to a daily frequency. We then calculate two commonly used performance metrics $\phi_{j,k}$ for every j -th rule, namely: the mean excess return ($\bar{f}_j$) and the Sharpe ratio (SR_j):

$$\bar{f}_j = \frac{1}{T} \sum_{t=1}^T (r_{j,t} - r_{b,t}) \quad (3a)$$

$$SR_j = \frac{\bar{r}_j}{\sigma_{r_j}} - \frac{\bar{r}_b}{\sigma_{r_b}} \quad (3b)$$

Where $r_{j,t}$ is the return of the j -th technical trading rule, $r_{b,t}$ is the return of the buy-and-hold benchmark, and σ_{R_j} , σ_{r_b} and σ_{f_j} are the standard deviation of the return of the j -th rule, the benchmark and the excess return respectively.

3.3. Outperformance and data mining

While testing our extensive trading rule universe, there is always a chance that we find highly profitable rules as a result of luck, rather than the forecasting ability of that particular rule (i.e. data mining). Various multiple hypothesis testing methods have been developed to mitigate this problem. These methods can generally be subdivided in 2 types, namely: controlling the Family wise error rate (FWER) or the false discovery rate (FDR).

The FWER is defined as the probability of rejecting at least one true null hypothesis. These methods are useful if you want to limit the probability of erroneously selecting at least one "lucky" technical trading rule. However, Romano et al. (2008) argue that controlling FWER can be too conservative, especially when the universe of rules is large. In practice, the strict control of the FWER is not necessary, as investors typically combine trading signals of multiple strategies.

As a result it is more beneficial for them to identify a large number of outperforming rules, while allowing a certain amount of "lucky" rules. This is exactly what the FDR does. It tries to control the expected proportion of falsely rejected null hypotheses (Benjamini and Hochberg, 1995). In this paper we will use the $FDR^{+/-}$ methodology introduced by Barras et al. (2010) and adapted for the estimation of technical trading rules by Bajgrowicz and Scaillet (2012). After computing the performance metrics ϕ_j for every rule j , $1 \leq j \leq J$, we test whether it has a statistically positive or negative performance compared to the benchmark:

$$H_{0,j} : \phi_j = 0; \quad H_{a,j} : \phi_j \neq 0 \quad (4)$$

Among the significant rules¹², we define R^+ as the amount of rules that have a positive performance, $\phi_j > 0$. Among these, F^+ is the number of significantly positive rules which were chosen by luck. The FDR among the rules yielding positive returns, denoted by FDR^+ , is defined as the expected proportion of the trading rules falsely determined as outperforming. The FDR^+ can be estimated as $\hat{FDR}^+ = \frac{\hat{F}^+}{\hat{R}^+}$, with $\hat{F}^+$ and $\hat{R}^+$ being estimators of F^+ and R^+ (see Appendix C). A FDR^+ of $k\%$ means that on average $k\%$ of the selected rules do not genuinely outperform the benchmark. As in Bajgrowicz and Scaillet (2012) we form $10\%-FDR^+$ portfolios, allowing for 10% not truly outperforming trading rules in our selection.

¹²In order to obtain the individual p-values for every rule, we use the stationary bootstrap from Politis and Romano (1994) with $q = 0.1$ and $N = 1000$.

3.4. Out-of-sample portfolios

Starting from January 2014 onwards, every month we determine the best performing rules of the past year and hold them in our portfolios until the end of the month. This results in a OOS period of 2,922 trading days (8 years). We construct 3 different types of portfolios, namely a portfolio containing: only the best performing rule, the 50 best performing rules, and a 10%-FDR⁺ portfolio, where the FDR⁺ level is fixed at 10%. An equal amount is invested in the selected technical trading rules at the start of the month and these position are held until the end of that month without rebalancing¹³. In order to construct the portfolios the logarithmic returns are first converted to normal returns, and afterwards the equally-weighted portfolio returns are converted back to logarithmic returns.

4. Results

Table 3 shows the OOS-performance of our in-or-out strategy portfolios after transaction costs and the performance of the buy-and-hold benchmark. Panel (a) shows the general performance of portfolios containing the best performing rule (Best), the 50 best performing rules (Best 50), and the rules selected by the FDR⁺-methodology (FDR-10%) in terms of either the mean excess return or the Sharpe ratio. Starting from the left, the total (logarithmic) return shows that the three mean-return portfolios perform slightly better, but not statistically significant, in absolute terms, than the benchmark portfolio, while the FDR⁺ portfolio based on the Sharpe ratio has a similar performance as the benchmark. The subdivision between positive and negative periods shows us that all rules decrease the size of the returns in both directions by about half. This could be an indication that the trading rules limit risk. The risk measures confirm that this is indeed the case. More specifically, the maximum drawdown (MDD) and the 95%-VaR (and the skewness) show that our portfolios are successful in decreasing downside risk. The SR portfolios have the lowest risk, paired with a lower return. As a result of the lower risk most portfolios also statistically significantly outperform the risk-return measures of the benchmark. All portfolios have a comparable amount of transactions between 450 and 750 (more than 55 per year) and average transaction costs of around -0.14%-points per trade. The break-even-costs (BEC) show us how high a %-point increase in the transaction costs can be until the strategy has a return of 0 (or equal to the benchmark between brackets). This ranges from 0.43% to 0.87%. Our outperforming rules only need an increase of less than 0.05%-points before breaking even with the benchmark. Lastly, the POR shows the average percentage of outperforming rules (in sample). We see that only 1.68% of the trading rules, on average, outperformed the benchmark in terms of mean excess return and 2.95% in terms of the Sharpe ratio.

Panel (b) shows the performance of the FDR-10% portfolios calculated for the individual trading frequencies. In terms of return and risk-adjusted measures, we see that intraday portfolios outperform the daily portfolios, showing us that there is indeed some value in trading on higher frequencies. Unexpectedly, risk-wise the portfolios perform similarly. The faster reactions of higher trading frequencies do not result decrease or increase in risk compared to the daily trading frequency. However, the higher trading frequencies execute a substantial larger amount of transactions. As a result the BEC of the daily portfolio is much higher, making their performance more robust against small increases of transaction costs or the bid-ask spread (similar to the findings of Svogun and Bazán-Palomino (2022)). Finally, the POR shows that the FDR-10% methodology finds much less outperforming rules for the 10 min. frequency. A possible reason is that the larger amount of transactions and costs makes more trading rules unprofitable.

Panel (c) shows the performance of the FDR-10% portfolios calculated for the individual classes of technical trading rules. Here we see that mainly the MA, Filter and the OBV rules for the mean portfolios and the OBV for the SR portfolios outperform all other rules in terms of absolute return measures and risk-return measures even though their POR is not much different than the other rules. The MA and OBV

¹³Elliott and Timmermann (2008) show that giving a larger weight to the performing best trading rules can decrease the influence of bad rules, especially when a large amount of rules are selected. Bajgrowicz and Scaillet (2012), however, do mention that giving a larger weight to the best rules has an effect very similar to reducing the FDR⁺ target level. Weighting the technical trading rules by their contribution to the total in-sample return of the portfolio did not change the OOS-performance significantly.

Portfolio	Return				Risk				Risk-Return		Trades and costs				POR
	Total	+	-		S.D.	Skew.	Kurt.	MDD	VaR	SR	Calmar	# Trades	Av. TC	BEC (Excess)	
Benchmark	4.15	40.48	-36.33		0.77	-0.81	11.87	-1.80	-6.34%	0.66	0.28	-	-	-	
Best	(a) General performance														
<i>Mean</i>	4.22	21.42	-17.21		0.52	0.48	9.76	-1.07	-4.02%	0.99	0.48	766.00	-0.14%	0.55% (0.01%)	-
<i>SR</i>	1.96	11.20	-9.23		0.40	0.70	24.12	-0.97	-2.17%	0.59	0.24	462.00	-0.13%	0.43% (-0.47%)	-
Best 50															
<i>Mean</i>	4.54	22.38	-17.84		0.47	0.91	9.01	-0.61	-3.47%	1.20**	0.91	723.70	-0.15%	0.63% (0.05%)	-
<i>SR</i>	2.89	12.31	-9.42		0.31	1.42	19.08	-0.45	-1.93%	1.13**	0.79	483.62	-0.14%	0.60% (-0.26%)	-
FDR-10%															
<i>Mean</i>	4.19	26.43	-22.24		0.56	0.11	8.14	-1.40	-4.09%	0.93*	0.37	483.46	-0.14%	0.87% (0.01%)	1.68%
<i>SR</i>	4.15	19.66	-15.51		0.41	1.10	11.18	-0.49	-2.82%	1.24**	1.04	694.10	-0.15%	0.60% (0.00%)	2.95%
Mean	(b) FDR-10% by frequency														
<i>Daily</i>	3.39	24.81	-21.42		0.58	-1.46	30.76	-1.02	-4.25%	0.71	0.41	55.62	-0.14%	6.10% (-1.35%)	1.82%
<i>60 Min.</i>	5.03	27.44	-22.41		0.58	-0.31	12.09	-0.98	-4.45%	1.07**	0.63	511.39	-0.16%	0.98% (0.17%)	2.57%
<i>30 Min.</i>	4.55	26.29	-21.73		0.55	0.49	7.17	-0.84	-4.20%	1.02*	0.66	789.86	-0.15%	0.58% (0.05%)	1.46%
<i>10 Min.</i>	5.04	26.82	-21.78		0.57	-0.02	9.98	-1.03	-4.15%	1.09**	0.60	608.89	-0.15%	0.83% (0.15%)	0.87%
SR															
<i>Daily</i>	3.54	17.35	-13.81		0.40	0.33	12.90	-0.68	-2.62%	1.08*	0.64	84.21	-0.15%	4.21% (-0.71%)	3.20%
<i>60 Min.</i>	4.17	20.19	-16.02		0.44	1.08	12.67	-0.61	-3.14%	1.18**	0.84	725.59	-0.14%	0.58% (0.00%)	4.91%
<i>30 Min.</i>	4.20	20.07	-15.87		0.44	1.23	11.89	-0.56	-2.89%	1.18**	0.93	958.99	-0.15%	0.44% (0.01%)	2.61%
<i>10 Min.</i>	3.34	21.46	-18.12		0.48	0.57	9.58	-0.65	-3.33%	0.85	0.63	1144.20	-0.14%	0.29% (-0.07%)	1.11%
Mean	(c) FDR-10% by technical trading rule														
<i>MA</i>	4.82	25.81	-20.99		0.57	0.21	8.12	-1.59	-4.49%	1.04*	0.37	290.60	-0.18%	1.66% (0.23%)	1.37%
<i>RSI</i>	2.09	16.54	-14.45		0.48	-2.99	65.72	-0.96	-2.85%	0.53	0.26	31.50	-0.14%	6.63% (-6.53%)	1.44%
<i>StoR</i>	3.11	21.64	-18.52		0.51	-0.09	10.10	-1.01	-3.92%	0.74	0.38	291.50	-0.15%	1.07% (-0.35%)	1.58%
<i>Filter</i>	4.63	23.68	-19.05		0.53	0.09	8.33	-0.75	-4.03%	1.08*	0.76	128.36	-0.15%	3.61% (0.38%)	2.22%
<i>CB</i>	3.48	23.77	-20.28		0.54	-0.08	7.85	-1.18	-4.38%	0.79	0.36	232.42	-0.14%	1.50% (-0.28%)	1.69%
<i>OBV</i>	4.56	28.42	-23.86		0.58	0.26	7.10	-0.76	-4.42%	0.97*	0.73	676.83	-0.16%	0.67% (0.06%)	2.20%
SR															
<i>MA</i>	3.38	20.80	-17.42		0.47	0.27	11.46	-0.75	-3.32%	0.88	0.55	599.44	-0.16%	0.56% (-0.13%)	3.17%
<i>RSI</i>	1.58	12.77	-11.18		0.43	-3.93	95.60	-0.64	-2.22%	0.44	0.29	48.82	-0.13%	3.24% (-5.25%)	1.05%
<i>StoR</i>	1.96	14.49	-12.53		0.38	0.41	14.30	-0.79	-2.59%	0.63	0.30	357.58	-0.15%	0.55% (-0.61%)	2.11%
<i>Filter</i>	2.80	14.35	-11.54		0.37	1.19	25.81	-0.52	-2.08%	0.92	0.65	227.81	-0.11%	1.23% (-0.59%)	2.73%
<i>CB</i>	1.70	14.19	-12.49		0.37	0.32	22.14	-0.60	-2.26%	0.56	0.34	218.79	-0.15%	0.78% (-1.12%)	1.85%
<i>OBV</i>	4.98	23.20	-18.21		0.47	1.03	9.32	-0.63	-3.25%	1.30***	0.98	947.37	-0.15%	0.53% (0.09%)	4.86%

Table 3: Out-of-sample performance (after transaction costs) for different in-or-out strategies. Four types of measures are shown for every strategy: return, risk, risk-return measures, and trades and costs. The return measures are the total (logarithmic) return, where a total return of $4.15 = 415\%$ (or $e^{4.15} - 1 = 6243.4\%$), and the total return during positive (+) and negative (-) periods in the benchmark. The risk measures are the annualized (ann.) standard deviation, skewness, kurtosis, maximum drawdown (MDD) and the 95% value-at-risk respectively. As risk-return measures we consider the ann. Sharpe (SR) and Calmar ratios. As trades and costs measures we show the amount of trades (#), the average cost of 1 transaction (Av. TC), and the break-even-transaction costs (BEC). The BEC shows how high a %-point increase in transaction costs can be until the total return of the strategy is 0 (or equal to the benchmark between brackets). Lastly, POR is the proportion of the trading rules, which were considered for the strategy, that are selected as statistically outperforming by the FDR-10% methodology. The table shows the performance of the Benchmark, Best, Best 50, FDR-10% portfolios, and the FDR-10% portfolios calculated for the individual trading frequencies and trading rule classes, for each performance measure. *, ** and *** indicate that designating this OOS-portfolio as statistically significantly outperforming the buy-and-hold benchmark, using bootstrap p-values, limits the FDR+ -level to 10, 5 and 1%, respectively.

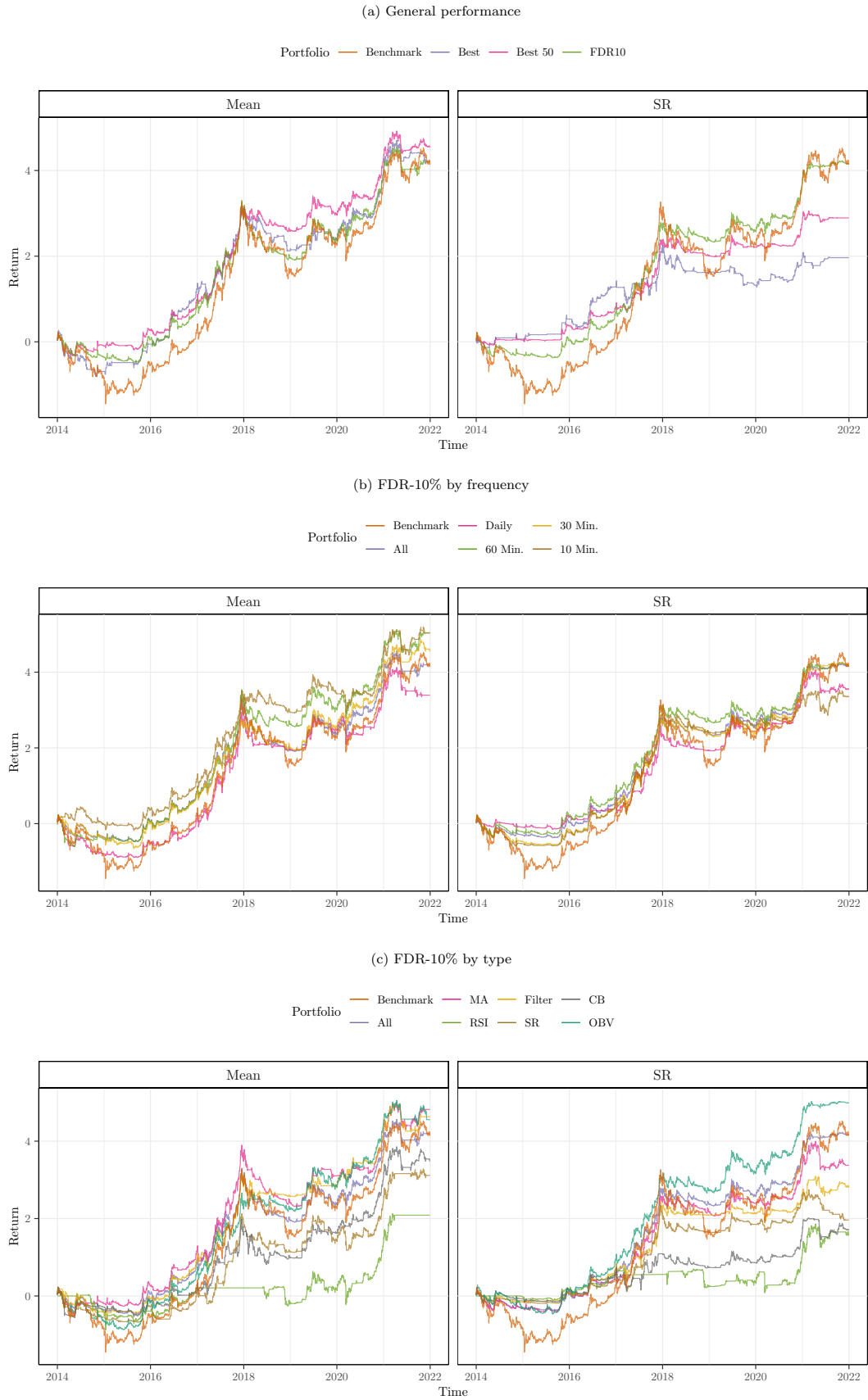


Figure 2: Out-of-sample cumulative return, after transaction costs, of our strategies against the benchmark over time. Panel (a) shows the cumulative returns (expressed in log returns) for both performance measures, for the benchmark portfolio, the portfolio consisting of only the best performing TTR (best), the portfolio consisting of the best 50 performing TTRs, and the FDR-10% portfolio. Panel (b) and (c) show the cumulative returns for the FDR-10% portfolios calculated for the unique trading frequencies and technical trading rule classes respectively.

Portfolio	2014-2015			2016-2017			2018-2019			2020-2021		
	Total	SR	POR	Total	SR	POR	Total	SR	POR	Total	SR	POR
Benchmark	-0.53	-0.36	-	3.47	2.33	-	-0.66	-0.45	-	1.86	1.14	-
(a) General performance												
Best												
<i>Mean</i>	-0.06	-0.07	-	3.01	2.58	-	-0.56	-0.61	-	1.82	1.58	-
<i>SR</i>	0.53	0.92	-	1.60	1.37	-	-0.80	-1.12	-	0.64	1.09	-
Best 50												
<i>Mean</i>	0.27	0.39	-	2.73	2.58	-	-0.03	-0.08	-	1.57	1.55	-
<i>SR</i>	0.37	0.88	-	1.90	2.32	-	-0.07	-0.16	-	0.68	1.46	-
FDR-10%												
<i>Mean</i>	0.05	0.06	3.76%	2.93	2.10	0.11%	-0.61	-0.68	2.78%	1.82	1.53	0.05%
<i>SR</i>	0.03	0.05	4.93%	2.62	2.63	0.66%	-0.07	-0.13	5.56%	1.56	1.99	0.65%
(b) FDR-10% by frequency												
Mean												
<i>Daily</i>	-0.50	-0.60	4.73%	3.13	2.19	0.02%	-0.15	-0.21	2.51%	0.92	0.65	0.01%
<i>60 Min.</i>	0.04	0.04	5.80%	3.18	2.60	0.10%	-0.19	-0.21	4.31%	2.00	1.48	0.06%
<i>30 Min.</i>	0.01	0.00	2.59%	2.58	2.02	0.16%	-0.05	-0.08	2.98%	2.01	1.77	0.13%
<i>10 Min.</i>	0.37	0.52	2.00%	2.91	2.24	0.18%	0.15	0.10	1.29%	1.61	1.20	0.02%
SR												
<i>Daily</i>	0.17	0.36	6.32%	2.15	2.35	0.31%	0.11	0.09	5.44%	1.10	1.17	0.73%
<i>60 Min.</i>	0.21	0.30	9.51%	2.79	2.68	1.39%	-0.25	-0.34	8.10%	1.43	1.65	0.65%
<i>30 Min.</i>	-0.18	-0.30	2.39%	2.80	2.53	0.70%	-0.23	-0.30	6.43%	1.81	2.16	0.93%
<i>10 Min.</i>	-0.24	-0.39	1.64%	3.04	2.52	0.20%	-0.28	-0.38	2.27%	0.83	0.78	0.32%
(c) FDR-10% by technical trading rule												
Mean												
<i>MA</i>	0.27	0.31	2.83%	3.30	2.35	0.01%	-0.47	-0.56	2.64%	1.71	1.33	0.01%
<i>RSI</i>	-0.39	-0.49	4.33%	0.60	0.88	0.04%	-0.04	-0.08	1.34%	1.92	1.51	0.04%
<i>S&R</i>	-0.27	-0.35	4.30%	2.12	1.92	0.01%	-0.41	-0.49	2.00%	1.68	1.36	0.02%
<i>Filter</i>	-0.06	-0.09	5.71%	2.97	2.35	0.02%	-0.07	-0.12	3.12%	1.79	1.52	0.01%
<i>CB</i>	-0.12	-0.15	4.60%	1.78	1.53	0.02%	-0.05	-0.08	2.07%	1.87	1.55	0.06%
<i>OBV</i>	-0.17	-0.17	3.04%	2.50	1.84	0.81%	0.50	0.43	4.60%	1.72	1.43	0.34%
SR												
<i>MA</i>	0.10	0.13	5.34%	2.36	2.30	0.32%	-0.06	-0.12	6.65%	0.98	0.92	0.36%
<i>RSI</i>	0.17	0.60	2.84%	0.38	0.57	0.04%	-0.20	-0.31	1.22%	1.23	0.91	0.11%
<i>S&R</i>	0.06	0.14	4.63%	1.99	2.00	0.24%	-0.20	-0.36	3.23%	0.11	0.13	0.35%
<i>Filter</i>	0.07	0.19	5.92%	2.12	2.05	0.18%	-0.07	-0.24	4.47%	0.67	0.75	0.33%
<i>CB</i>	0.15	0.34	3.51%	0.94	0.92	0.27%	-0.23	-0.44	3.08%	0.84	1.12	0.56%
<i>OBV</i>	0.13	0.16	4.83%	2.90	2.79	3.18%	0.15	0.10	8.95%	1.80	2.04	2.47%

Table 4: Out-of-sample performance (after transaction costs) for different in-or-out strategies during different sub-periods. For every sub-period we show the total (logarithmic) return, the ann. Sharpe (SR) ratio and the POR, the average proportion of the trading rules, which were considered for the portfolio, that are selected as statistically outperforming by the FDR-10% methodology. The table shows the performance of the Benchmark, Best, Best 50, FDR-10% portfolios, and the FDR-10% portfolios calculated for the individual trading frequencies and trading rule classes, for each performance measure. *, ** and *** indicate that designating this OOS-portfolio as statistically significantly outperforming the buy-and-hold benchmark, using bootstrap p-values, limits the FDR⁺-level to 10, 5 and 1%, respectively.

rules also have a larger amount of trades than the other rules. The outperformance of the OBV rules may indicate that volume is important in predicting bitcoin prices. The data in our sample indeed shows that 70% of the trade volume took place on the days with the 30% highest absolute return. The RSI rules are generally the worst performers, mainly because the best performing RSI rules do not execute a lot of trades, and as a result the RSI portfolio is not always in the market during bull runs (see figure 2).

Figure 2 show the cumulative returns of all these portfolios over time. Panel (a) show the cumulative returns for the general portfolios. It shows clearly that our trading rules manage to limit losses during the downturns in 2014-2015 and 2018-2019. The mean-return portfolios also can predict the bullish periods well, which results into a similar total return, with less risk compared to the benchmark. Panel (b) shows the cumulative returns of the FDR-10% portfolios calculated for the individual trading frequencies. Here we see that the portfolios trading on a daily frequency react slower to the bullish periods in 2016-2017

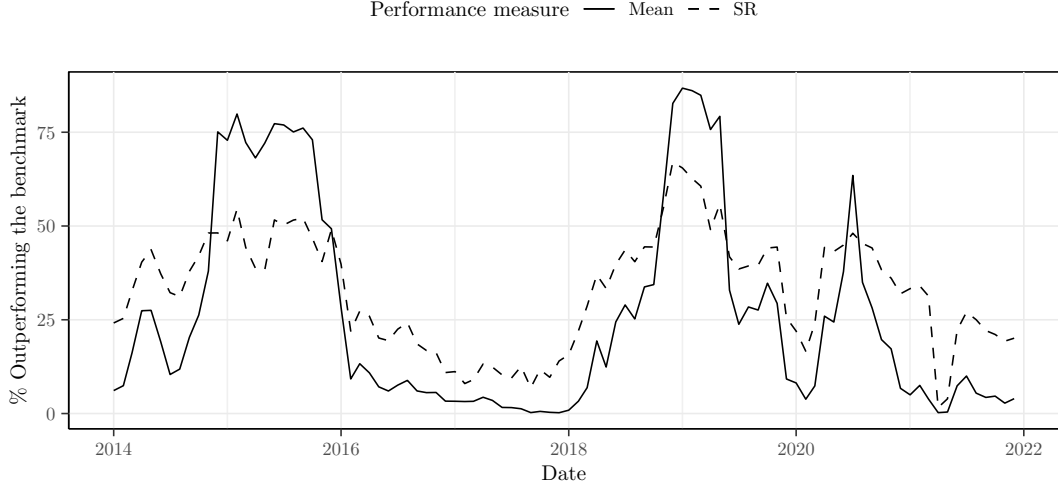


Figure 3: The % of trading rules with a better performance than the buy-and-hold benchmark during our sample in terms of the mean and Sharpe ratio (SR).

and 2020-2021, leading them to have a lower overall return. Panel (c) shows the cumulative return of the FDR-10% portfolios calculated for the individual classes of technical trading rules. Most trading rules limit the downside risk during the downturns. The outperformance of the OBV rules can be clearly seen. Lastly, we note the bad performance of the RSI rules. As a trend reversal trading rule it is seemingly not able to predict the bull market in 2017, and stays out of the market for this whole period (which results in a flat cumulative return for the Mean RSI portfolio), which results in an underperformance in comparison to the benchmark and other trading rules.

Table 4 shows the performance of our portfolios during 4 different 2-year sub-periods between 2014, 2016, 2018, 2020 and 2022. The first and third sub-periods are marked by a significant downturn. During these periods all portfolios outperformed the benchmark, albeit not statistically significant¹⁴. This confirms what the MDD and 95%-VaR statistics in table 3 have shown us: the technical trading portfolios are able to limit the downside risk. Interestingly, while some portfolios still managed to turn a profit in 2014-2015, most only managed to limit their losses during 2018-2019. This may suggest that the bitcoin market was more efficient during this period, which is in line with some recent literature suggesting that the bitcoin market is becoming more efficient over time (Nadarajah and Chu, 2017; Bariviera et al., 2017; Sensoy, 2019). The second and fourth sub-period are marked by a significant bull run. Here we see that most portfolios at best manage to have a similar performance to the benchmark, both in total return and in terms of the Sharpe ratio. Some portfolios manage to slightly, but not significantly, outperform the buy-and-hold benchmark in terms of the Sharpe ratio. The POR also shows that the SR portfolios manage to find more outperforming rules during this period, although in a much smaller scale than in the first and third period. This can also be seen in figure 3. The figure shows the % of rules with a better performance than the buy-and-hold benchmark during each selection period (here a duration of 1 year). We clearly see 3 spikes of a large amount of outperforming rules during the downturns in 2014-2015, 2018 and 2020. This is in line with the findings of Svogun and Bazán-Palomino (2022), that trading rules do not outperform the buy-and-hold benchmark during bubbles.

Lastly in table 4, in bold are the best performing portfolios in every period for every measure and panel separately. The best performing frequencies, in the subsequent sub-periods are the 10, 60, 10 and 30 minute portfolios have the best performances. Compared to the first 3 periods, the daily and 10 min.

¹⁴Some portfolios are statistically significant using individual tests, however after correcting for data mining the significance disappears.

period underperformed the other portfolios in 2020-2021. For the technical trading rules, the MA, MA, OBV and RSI portfolios had the best absolute performance, and the RSI and 3 times the OBV portfolios had the best risk-adjusted performance. Unexpectedly, the RSI portfolio was the best performing rule during 2 sub-periods, even though it has the worst performance over the complete sample. The first and fourth period also have completely different characteristics, making it hard to link the outperformance to market conditions. The OBV performed well during all periods, being the only rule that was able to always generate a profit. As the only rule that takes volume in consideration, this may be an interesting topic for future research. If we compare the best performing portfolios over the different periods, we can only conclude that there is no observable pattern. Different types of portfolios, based on different performance measures, trading frequencies, trading rules and selection procedures outperform or underperform in every period. This observation is reconcilable with the adaptive market hypothesis of Lo (2004), and arbitrage theory. The market moves in waves of efficiency, possible opportunities for profit can be exploited for some time, however not continuously.

5. Discussion

Although we show that it is possible to outperform, in terms of absolute return or risk-adjusted performance measures, a buy-and-hold strategy in bitcoin using simple technical trading rules, certain remarks about our results can be made.

First, because we do not allow shorting, our technical trading rules may not be able to profit overtly from price decreases. Shorting is not possible on Bitstamp, but may have been possible on other bitcoin exchanges during our sample. Usually shorting has extra costs in form of a borrow rate. For our results we used a borrow rate of 0.02% for every 4 hours the position is opened¹⁵. In table A.5 in Appendix A, panel (a), we show the general performances of our portfolios with shorting allowed¹⁶. We see that in comparison with the in-or-out strategies most strategies have higher absolute returns, paired with an increase of the (downside) risk. As a result most portfolios also outperform the benchmark in terms of risk-return measures, indicating that our technical trading rules indeed can profit from price decreases. However, the extra borrow rate results in a increase of the transaction costs, tempering the increase of the absolute returns, making the risk-return measures comparable to the portfolios of our in-or-out strategies. Contrarely to the in-or-out strategies, none of the performance measures are statistically significant. Panel (b) shows the subdivision by trading frequency. Here we see that intraday frequencies (mainly the 30 min.) still perform significantly better than the daily frequency, showing that intraday portfolios profit from shorting. The transaction costs show us that the daily portfolios also suffer from higher carrying costs for daily short positions, reducing their overall return. panel (c) shows the subdivision by technical trading rule. as was the case for the in-or-out strategy, the OBV rules outperform the other technical trading rules. The MA and Filter rules still perform well, however the difference with the OBV rules has increased. Lastly, the risk measures of the general FDR-10% and Best 50 portfolios are generally lower than those of the other portfolios, indicating that there may be some diversification benefits.

Second, in our approximation of realistic transaction costs, we use the lowest possible trading fee and bid-ask spread. Taking into account that the transaction fee on Bitstamp is volume based¹⁷, in reality there should be a trade-off between the transaction fees and the bid-ask spread, where a low(high) transaction volume leads to high(low) fees per transaction, but a low(high) bid-ask spread. Knowing that there can be a difference of up to 0.50%-points in transaction fees and of 1%-points on average between the 0-BTC and 100-BTC-deep bid-ask spread, this could have a significant impact on our result, especially after seeing the BEC in table 3. Table A.6 in Appendix A shows the general performance of the in-or-out and shorting-allowed portfolios recalculated for the 2 extreme cases. Namely, the case with the highest¹⁸ bid-ask spread and the lowest transaction fees, and the case with the lowest bid-ask-spread and the highest transaction fees.

¹⁵This is the borrow rate/margin fee currently used on the Kraken.com exchange.

¹⁶With $D_{j,t-1} \in \{1, 0, -1\}$.

¹⁷A higher trading volume results in lower transaction fees and vice versa.

¹⁸The 100-BTC-deep bid-ask spread is the deepest spread that was available to us.

Looking at the average transaction costs, we see that they more than quintuple due to a higher bid-ask and triple due higher fees. As a result all selected outperforming rules execute markedly less trades. Because of the higher costs, our portfolios performed worse than in our ideal case, however most of the in-or-out and some of the shorting-allowed portfolios still managed to outperform the benchmark risk-return wise. The interpretation of the risk measures also has not changed, although they all became insignificant. Surprisingly, the lower amount of trades, leads to a much higher BEC for all our portfolios, making their performance more robust to increases in transaction costs. What is not shown in the table is that because of the higher transaction costs, the portfolios trading on daily frequencies now perform comparable or better than the intraday portfolios. The trading rules with a higher amount of trades (e.g. MA and OBV) also take a larger hit to their performance.

Third, we know from theory that in order to avoid data mining when selecting outperforming technical trading rules, we should construct a large enough technical trading rule universe. However there is a trade-off, as using too many technical trading rules can also decrease the power of our statistical tests. In our study we used a substantial amount of 75,360 technical trading rules, which could have influenced our ability to select truly outperforming rules. Table A.7 in Appendix A, panel (a), replicates the general results of table 3 using only 1/6th of the technical trading rules. This amounts to 12,340 technical trading rules analysed, which is more comparable to the amount of technical trading rules used in other studies about technical trading rule performance. However, looking at the table we see that using a smaller investment universe produces very similar results.

Fourth, every month we select the best performing technical trading rules of the past year in order to form our portfolios. An argument can be made, however, that a shorter forming period will increase profits, as the effectiveness of specific technical trading rules is time-varying. panel (b) and (c) of table 3 shows the general performance of portfolios formed with a 6- and 3-month forming period respectively. We see that most of these portfolios still significantly outperform the benchmark risk-return wise, and generally underperform the 12-month forming period portfolios. The longer forming period probably results in a more stable prediction for finding truly outperforming trading rules in the volatile BTC market.

Fifth, although it is not possible on Bitstamp, margin trading could also be an alternative to the shorting-allowed and in-or-out strategy. Our margin trading strategy is as follows: if a neutral signal is given, go long in bitcoin. If a buy signal is given, borrow money at a fixed fee and double your position, on the other hand if a sell signal is given, liquidate the position and buy the risk-free asset¹⁹. The results are shown in panel (d) from table A.7 in Appendix A. We see that all portfolios largely, and now significantly, outperform all earlier discussed portfolios in terms of absolute returns. These large returns are however paired with an increased risk. Interestingly, although the margin trading strategies still outperform the benchmark risk-return wise, the increased risk results in most of the in-or-out and shorting-allowed strategies having a better risk-adjusted performance.

Finally, in our sample, we exclusively use data from Bitstamp. As mentioned before, this has some benefits: having a more uniform dataset, more precisely being able to estimate transaction costs and being able to take advantage from the longevity and trustworthiness of the exchange, possibly diminishing the impact of time-varying risk and liquidity premia linked to the exchange. Research has shown that significant differences in price forming between exchanges can exist (Szetela et al., 2020; Blasco et al., 2022). First, keeping in mind that Bitstamp does not allow short, margin and derivative trading, it is possible that some of the performance of our portfolios is a result of a slower reaction of the exchange prices to news. We also find that trading on an intradaily frequency performs better than trading on a daily frequency, which may support this hypothesis. However, given the relatively large daily volume on the exchange, price deviations should in theory be arbitrated away relatively quickly. Second, as Bitstamp is an European exchange, logically most of the volume takes place during European convenient times. As a result some questions about the scalability of our trading strategies can be made, especially when trades take place during the night.

¹⁹With $D_{j,t-1} \in \{2, 1, 0\}$. We set the margin trading fee at 0.02% for every 4 hours the position is opened, as was the case with the borrow rate in the shorting-allowed portfolios.

6. Conclusion

In this paper we study if the use of simple technical trading rules can outperform buying and holding bitcoins. We attempt to do this by first selecting outperforming rules, from a set of 75,360 possible trading rules, then combining them in different portfolios and finally assessing their performance out-of-sample after 'realistic' transaction costs. Compared to earlier research, which generally concludes that trading rules can outperform a buy-and-hold strategy in the bitcoin market, we apply much more restrictive conditions (transaction costs, out-of-sample performance, data mining corrections) and search over a higher amount of technical trading rule classes, parameterizations and trading frequencies. Unlike in Hudson and Urquhart (2021), we find that our rules still can significantly outperform the buy-and-hold strategy out-of-sample, mainly risk-return wise. More specifically, as in Svogun and Bazán-Palomino (2022), we find that technical trading rules can be very successful in limiting downside risk and large drawdowns, while not giving additional returns during bubbles. However, we do not find the significant outperformance in absolute terms that was found in the studies that did not consider transaction costs Corbet et al. (2019); Miller et al. (2019); Grobys et al. (2020); Detzel et al. (2021), showing the importance of taking costs into consideration. Additionally we find that combining multiple technical trading rules, across rule class and trading frequency, has diversification benefits. Although trading on higher frequencies usually increases returns, it makes the portfolio more vulnerable to small and unexpected increases in transaction costs. Trading on daily frequency is much more robust to these increases. We also find that mainly the on-balance volume and in a smaller manner the moving average and filter rules (which also performed well in earlier studies) performed better than the other technical trading rules. However, when we look at the different sub-periods, we see that the performance of various trading rules and on various trading frequencies is time-variant, which may explain the differences in best performing trading rules found in earlier research. The time-variation of the outperforming trading rules is in line with the adaptive market hypothesis of Lo (2004) and arbitrage theory, showing that the bitcoin market, at least during specific times is not efficient. Lastly, we find that allowing shorting and margin trading increases performance, however, this is paired with an increase in (downside) risk and transaction costs.

Appendix A. Extra tables

Portfolio	Return		Risk			Risk-Return		Trades and costs			POR		
	Total	+	-	S.D.	Skew.	Kurt.	MDD	Var	SR	Calmar		# Trades	Av. TC
Benchmark	4.15	40.48	-36.33	0.77	-0.81	11.87	-1.80	-6.34%	0.66	0.28	-	-	-
(a) General performance													
Best													
Mean	4.52	6.97	-2.44	0.75	0.45	9.63	-1.23	-5.82%	0.74	0.45	1027.00	-0.27%	0.44% (0.04%)
SR	1.94	6.32	-4.38	0.60	0.84	21.82	-1.19	-4.30%	0.39	0.20	1028.00	-0.19%	0.19% (-0.21%)
Best 50													
Mean	5.31	10.41	-5.10	0.56	1.31	14.58	-0.56	-3.74%	1.18	1.17	1217.62	-0.26%	0.44% (0.10%)
SR	3.96	7.01	-3.05	0.42	1.93	21.04	-0.44	-2.59%	1.16	1.10	1032.88	-0.22%	0.38% (-0.02%)
FDR-10%													
Mean	4.58	17.59	-13.01	0.62	0.24	5.93	-1.07	-4.60%	0.91	0.52	802.44	-0.26%	0.57% (0.05%)
SR	4.25	13.39	-9.13	0.49	0.76	10.19	-0.78	-3.34%	1.07	0.67	852.67	-0.22%	0.50% (0.01%)
(b) FDR-10% by frequency													
Mean													
Daily	2.91	13.42	-10.51	0.67	-0.82	18.39	-1.53	-5.46%	0.53	0.23	79.30	-1.37%	3.68% (-1.55%)
60 Min.	6.08	17.93	-11.84	0.66	0.18	5.29	-0.81	-4.84%	1.14	0.93	788.37	-0.28%	0.77% (0.25%)
30 Min.	7.21	17.48	-10.27	0.65	0.76	7.92	-0.47	-4.63%	1.37	1.91	1235.51	-0.21%	0.58% (0.25%)
10 Min.	6.50	19.19	-12.69	0.67	0.34	4.96	-0.69	-5.05%	1.20	1.17	924.93	-0.27%	0.70% (0.26%)
SR													
Daily	1.98	6.85	-4.86	0.41	0.48	21.41	-0.71	-2.22%	0.58	0.33	97.81	-0.57%	2.02% (-2.21%)
60 Min.	4.17	12.90	-8.73	0.56	0.37	8.51	-0.85	-4.03%	0.92	0.60	847.31	-0.24%	0.49% (0.00%)
30 Min.	5.90	13.77	-7.87	0.56	1.00	9.72	-0.54	-3.95%	1.29	1.34	1186.50	-0.20%	0.50% (0.15%)
10 Min.	4.82	14.11	-9.29	0.63	-0.21	10.14	-0.71	-4.39%	0.94	0.83	1019.85	-0.26%	0.47% (0.07%)
(c) FDR-10% by technical trading rule													
Mean													
MA	5.50	14.13	-8.63	0.73	0.95	14.05	-0.91	-5.49%	0.93	0.74	695.36	-0.37%	0.79% (0.19%)
RSI	1.39	9.99	-8.60	0.54	-1.75	40.69	-1.32	-3.89%	0.30	0.12	39.42	-1.58%	3.53% (-6.98%)
SE/R	1.41	6.42	-5.01	0.69	0.96	16.49	-1.53	-5.27%	0.24	0.11	491.99	-0.41%	0.29% (-0.56%)
Filter	4.55	13.30	-8.75	0.65	1.13	20.83	-1.24	-4.83%	0.87	0.45	158.95	-0.66%	2.86% (0.26%)
CB	2.15	11.90	-9.74	0.70	0.24	19.41	-1.38	-5.51%	0.37	0.19	207.97	-0.66%	1.04% (-0.96%)
OBV	6.37	18.21	-11.84	0.65	0.29	6.35	-0.97	-4.80%	1.21	0.81	962.98	-0.25%	0.66% (0.23%)
SR													
MA	4.19	8.09	-3.89	0.65	0.27	7.51	-1.01	-4.79%	0.79	0.51	763.69	-0.35%	0.55% (0.01%)
RSI	0.14	7.23	-7.09	0.43	-3.47	90.22	-1.66	-2.69%	0.02	0.01	62.52	-0.61%	0.22% (-6.41%)
SE/R	3.32	3.84	-0.52	0.48	3.40	62.74	-0.62	-2.91%	0.85	0.66	448.98	-0.26%	0.74% (-0.18%)
Filter	2.92	9.52	-6.60	0.45	-0.21	18.27	-1.17	-2.56%	0.78	0.30	165.32	-0.36%	1.77% (-0.74%)
CB	0.77	5.09	-4.32	0.46	0.36	14.31	-0.79	-3.49%	0.19	0.11	338.08	-0.29%	0.23% (-1.00%)
OBV	6.37	16.33	-9.97	0.59	0.50	8.72	-0.71	-4.01%	1.32	1.11	1173.20	-0.22%	0.54% (0.19%)

Table A.5: Out-of-sample performance (after transaction costs) for the portfolios where shorting is allowed. Four types of measures are shown for every strategy: return, risk, risk-return measures, and trades and costs. The return measures are the total (logarithmic) return, where a total return of $4.15 = 415\%$ (or $e^{4.15} - 1 = 6243.4\%$), and the total return during positive (+) and negative (-) periods in the benchmark. The risk measures are the annualized (ann.) standard deviation, skewness, kurtosis, maximum drawdown (MDD) and the 95% value-at-risk respectively. As risk-return measures we consider the ann. Sharpe (SR) and Calmar ratios. As trades and costs measures we show the amount of trades (#), the average cost of 1 transaction (Av. TC), and the break-even-transaction costs (BEC). The BEC shows how high a %-point increase in transaction costs can be until the total return of the strategy is 0 (or equal to the benchmark between brackets). Lastly, POR is the proportion of the trading rules, which were considered for the strategy, that are selected as statistically outperforming by the FDR-10% methodology. The table shows the performance of the Benchmark, Best, Best 50, FDR-10% portfolios, and the FDR-10% portfolios calculated for the individual trading frequencies and trading rule classes, for each performance measure. *, ** and *** indicate that designating this OOS-portfolio as statistically significantly outperforming the buy-and-hold benchmark, using bootstrap p-values, limits the FDR+ level to 10, 5 and 1%, respectively.

Portfolio	Return			Risk				Risk-Return		Trades and costs				POR
	Total	+	-	S.D.	Skew.	Kurt.	MDD	VaR	SR	Calmar	# Trades	Av. TC	BEC (Excess)	
Benchmark	4.15	40.48	-36.33	0.77	-0.81	11.87	-1.80	-6.34%	0.66	0.28	-	-	-	-
(a) in-or-out, 100-BTC deep Bid-Ask, lowest fees														
Best														
Mean	3.31	23.85	-20.55	0.56	-0.45	9.28	-1.34	-4.60%	0.72	0.30	108.00	-0.98%	3.06% (-0.78%)	-
SR	1.87	12.47	-10.60	0.43	-0.06	28.63	-0.94	-2.41%	0.52	0.24	85.00	-0.71%	2.21% (-2.67%)	-
Best 50														
Mean	3.47	22.70	-19.24	0.49	-0.02	8.59	-0.91	-3.98%	0.86	0.47	111.32	-0.85%	3.12% (-0.61%)	-
SR	2.77	13.57	-10.80	0.33	0.61	16.69	-0.60	-2.09%	1.01	0.56	74.08	-0.78%	3.74% (-1.86%)	-
FDR-10%														
Mean	3.67	27.72	-24.05	0.61	-1.28	26.03	-1.14	-4.58%	0.74	0.39	120.76	-0.93%	3.04% (-0.39%)	0.78%
SR	2.72	19.14	-16.41	0.43	0.04	10.48	-0.78	-3.14%	0.76	0.42	130.73	-0.77%	2.08% (-1.09%)	0.95%
(b) in-or-out, 0-BTC deep Bid-Ask, highest fees														
Best														
Mean	3.33	23.37	-20.04	0.57	0.00	10.04	-1.28	-4.83%	0.72	0.32	273.00	-0.40%	1.22% (-0.30%)	-
SR	1.45	12.63	-11.17	0.42	-0.12	18.30	-0.78	-2.70%	0.41	0.22	208.00	-0.38%	0.70% (-1.29%)	-
Best 50														
Mean	3.63	22.59	-18.96	0.48	0.34	8.23	-0.85	-3.78%	0.92	0.52	274.82	-0.39%	1.32% (-0.19%)	-
SR	2.84	13.28	-10.44	0.32	0.93	17.40	-0.54	-1.95%	1.07	0.64	137.90	-0.39%	2.06% (-0.95%)	-
FDR-10%														
Mean	3.85	27.19	-23.34	0.58	-0.39	12.07	-0.89	-4.58%	0.81	0.53	218.54	-0.38%	1.76% (-0.13%)	1.11%
SR	2.95	19.95	-17.00	0.44	0.42	10.00	-0.70	-3.30%	0.83	0.52	306.23	-0.38%	0.96% (-0.39%)	1.62%
(c) shorting-allowed, 100-BTC deep Bid-Ask, lowest fees														
Best														
Mean	2.71	11.29	-8.58	0.75	-0.14	5.13	-1.83	-6.31%	0.44	0.18	179.00	-1.64%	1.51% (-0.80%)	-
SR	-0.94	3.82	-4.76	0.53	-0.26	13.46	-2.24	-4.40%	-0.24	-0.06	193.00	-1.18%	-0.49% (-2.64%)	-
Best 50														
Mean	1.77	10.34	-8.57	0.51	0.47	7.74	-0.95	-3.94%	0.41	0.22	266.78	-1.26%	0.66% (-0.89%)	-
SR	1.79	8.00	-6.21	0.37	0.91	14.70	-0.57	-2.67%	0.58	0.38	157.88	-1.18%	1.13% (-1.49%)	-
FDR-10%														
Mean	5.21	20.34	-15.12	0.63	-0.14	8.53	-1.03	-4.97%	1.02	0.62	176.75	-1.22%	2.95% (0.60%)	0.55%
SR	2.71	14.17	-11.45	0.47	0.20	13.94	-0.74	-3.24%	0.70	0.45	210.91	-0.99%	1.29% (-0.68%)	0.54%
(d) shorting-allowed, 0-BTC deep Bid-Ask, highest fees														
Best														
Mean	2.27	5.60	-3.33	0.76	-0.10	5.65	-1.88	-6.10%	0.36	0.15	440.00	-0.69%	0.52% (-0.43%)	-
SR	-0.98	4.60	-5.58	0.56	-0.19	11.85	-2.10	-4.51%	-0.23	-0.06	340.00	-0.59%	-0.29% (-1.51%)	-
Best 50														
Mean	3.12	9.87	-6.74	0.52	0.70	7.56	-0.71	-3.81%	0.73	0.54	631.74	-0.58%	0.49% (-0.16%)	-
SR	2.75	7.11	-4.36	0.38	1.44	16.55	-0.42	-2.61%	0.89	0.79	458.12	-0.51%	0.60% (-0.31%)	-
FDR-10%														
Mean	3.95	17.26	-13.30	0.64	0.02	6.97	-1.17	-4.98%	0.76	0.41	365.56	-0.61%	1.08% (-0.05%)	0.78%
SR	2.43	12.85	-10.42	0.47	0.52	12.40	-0.98	-3.42%	0.62	0.30	422.10	-0.52%	0.58% (-0.41%)	0.83%

Table A.6: Out-of-sample performance (after transaction costs) for the in-or-out and shorting-allowed strategies calculated with either a 100-BTC deep bid-ask spread or the highest possible trading fees on the platform. Four types of measures are shown for every strategy: return, risk, risk-return measures, and trades and costs. The return measures are the total (logarithmic) return, where a total return of 4.15 = 415% (or $e^{4.15} - 1 = 6243.4\%$), and the total return during positive (+) and negative (-) periods in the benchmark. The risk measures are the annualized (ann.) standard deviation, skewness, kurtosis, maximum drawdown (MDD) and the 95% value-at-risk respectively. As risk-return measures we consider the ann. Sharpe (SR) and Calmar ratios. As trades and costs measures we show the amount of trades (#), the average cost of 1 transaction (Av. TC), and the break-even-transaction costs (BEC). The BEC shows how high a %-point increase in transaction costs can be until the total return of the strategy is 0 (or equal to the benchmark between brackets). Lastly, POR is the proportion of the trading rules, which were considered for the strategy, that are selected as statistically outperforming by the FDR-10% methodology. The table shows the performance of the Benchmark, Best, Best 50, FDR-10% portfolios. *, ** and *** indicate that designating this OOS-portfolio as statistically significantly outperforming the buy-and-hold benchmark, using bootstrap p-values, limits the FDR⁺-level to 10, 5 and 1%, respectively.

Portfolio	Return			Risk				Risk-Return		Trades and costs				POR
	Total	+	-	S.D.	Skew.	Kurt.	MDD	VaR	SR	Calmar	# Trades	Av. TC	BEC (Excess)	
Benchmark	4.15	40.48	-36.33	0.77	-0.81	11.87	-1.80	-6.34%	0.66	0.28	-	-	-	-
(a) in-or-out, 1/6th investment universe														
Best														
Mean	3.95	20.70	-16.74	0.51	0.54	9.08	-1.10	-3.88%	0.96	0.44	582.00	-0.14%	0.68% (-0.03%)	-
SR	2.01	8.89	-6.88	0.35	1.46	32.62	-0.63	-1.70%	0.69	0.38	474.00	-0.13%	0.42% (-0.45%)	-
Best 50														
Mean	4.37	23.32	-18.95	0.48	0.71	7.99	-0.75	-3.58%	1.12**	0.72	637.54	-0.16%	0.69% (0.04%)	-
SR	3.36	15.36	-12.00	0.36	0.90	14.66	-0.44	-2.26%	1.15**	0.94	528.02	-0.14%	0.64% (-0.15%)	-
FDR-10%														
Mean	3.64	25.28	-21.64	0.55	0.06	7.82	-1.10	-4.11%	0.81	0.41	556.44	-0.13%	0.65% (-0.09%)	1.88%
SR	4.19	20.32	-16.13	0.43	1.11	10.93	-0.53	-2.98%	1.21**	0.96	733.48	-0.14%	0.57% (0.01%)	3.14%
(b) in-or-out, 6-month forming period														
Best														
Mean	4.64	19.36	-14.71	0.50	-0.27	16.01	-0.76	-3.36%	1.15*	0.75	930.00	-0.13%	0.50% (0.05%)	-
SR	2.57	10.02	-7.45	0.37	1.29	31.44	-1.24	-1.99%	0.84	0.25	691.00	-0.13%	0.37% (-0.23%)	-
Best 50														
Mean	3.68	20.58	-16.91	0.42	0.91	9.01	-0.76	-3.09%	1.06*	0.59	925.30	-0.13%	0.40% (-0.05%)	-
SR	2.29	11.35	-9.06	0.29	1.60	19.34	-0.43	-1.86%	0.96	0.64	605.56	-0.13%	0.38% (-0.31%)	-
FDR-10%														
Mean	3.31	26.50	-23.19	0.55	-0.28	8.93	-1.13	-4.06%	0.74	0.36	558.06	-0.13%	0.59% (-0.15%)	1.78%
SR	3.94	19.13	-15.18	0.40	1.25	11.54	-0.57	-2.78%	1.21*	0.85	761.60	-0.14%	0.52% (-0.03%)	2.55%
(c) in-or-out, 3-month forming period														
Best														
Mean	3.25	19.00	-15.75	0.51	-0.28	13.93	-0.91	-3.85%	0.78	0.44	1410.00	-0.11%	0.23% (-0.06%)	-
SR	1.54	11.92	-10.38	0.44	-3.37	85.10	-1.98	-2.73%	0.42	0.09	1028.00	-0.11%	0.15% (-0.25%)	-
Best 50														
Mean	3.84	20.72	-16.88	0.43	0.77	7.55	-0.79	-3.22%	1.10*	0.59	1018.38	-0.13%	0.38% (-0.03%)	-
SR	2.10	11.90	-9.80	0.29	0.70	14.91	-0.44	-1.97%	0.86	0.57	767.68	-0.12%	0.27% (-0.27%)	-
FDR-10%														
Mean	4.09	26.25	-22.16	0.51	0.11	6.17	-1.12	-3.99%	0.98	0.45	670.19	-0.13%	0.61% (-0.01%)	2.19%
SR	4.05	18.50	-14.46	0.38	1.30	11.50	-0.58	-2.52%	1.30**	0.86	815.39	-0.13%	0.50% (-0.01%)	2.88%
(d) margin trading strategy														
Best														
Mean	8.94***	47.62	-38.68	1.09	0.17	9.66	-2.04	-8.28%	1.01**	0.54	1075.00	-0.33%	0.83% (0.45%)	-
SR	5.21**	39.42	-34.20	1.03	0.43	15.13	-1.97	-7.58%	0.62	0.33	810.00	-0.33%	0.64% (0.13%)	-
Best 50														
Mean	8.36***	48.13	-39.76	0.98	0.91	8.23	-1.42	-7.10%	1.05**	0.73	1289.02	-0.31%	0.65% (0.33%)	-
SR	7.17***	43.53	-36.36	0.92	1.03	10.10	-1.14	-6.62%	0.97**	0.78	1080.10	-0.30%	0.66% (0.28%)	-
FDR-10%														
Mean	6.60***	46.00	-39.40	0.89	0.72	6.91	-1.52	-6.87%	0.91**	0.54	789.46	-0.34%	0.84% (0.31%)	6.77%
SR	5.89***	42.32	-36.43	0.82	0.63	7.63	-1.15	-6.27%	0.88**	0.63	767.95	-0.29%	0.77% (0.23%)	2.17%

Table A.7: Out-of-sample performance (after transaction costs) for the in-or-out strategy (a) calculated for a investment universe consisting of only 12,340 trading rules (1/6th of the original amount), (b,c) where the outperforming rules of the past (b) 6-months/(c) quarter) are selected, (d) and for the margin trading strategy. Four types of measures are shown for every strategy: return, risk, risk-return measures, and trades and costs. The return measures are the total (logarithmic) return, where a total return of 4.15 = 415% (or $e^{4.15} - 1 = 6243.4\%$), and the total return during positive (+) and negative (-) periods in the benchmark. The risk measures are the annualized (ann.) standard deviation, skewness, kurtosis, maximum drawdown (MDD) and the 95% value-at-risk respectively. As risk-return measures we consider the ann. Sharpe (SR) and Calmar ratios. As trades and costs measures we show the amount of trades (#), the average cost of 1 transaction (Av. TC), and the break-even-transaction costs (BEC). The BEC shows how high a %-point increase in transaction costs can be until the total return of the strategy is 0 (or equal to the benchmark between brackets). Lastly, POR is the proportion of the trading rules, which were considered for the strategy, that are selected as statistically outperforming by the FDR-10% methodology. The table shows the performance of the Benchmark, Best, Best 50, FDR-10% portfolios. *, **, and *** indicate that designating this OOS-portfolio as statistically significantly outperforming the buy-and-hold benchmark, using bootstrap p-values, limits the FDR+ -level to 10, 5 and 1%, respectively.

Appendix B. Technical Trading Rules

We compare a total of 18,840 technical trading rules divided over 6 commonly used classes and do this for 4 sampling frequencies, summing to 75,360 technical trading rules in total. In the following sections we will shortly²⁰ describe how we implement each of these classes of technical trading rules. Table B.8 in section Appendix B.7 shows all the used parameters for every calculated technical trading rule.

Appendix B.1. Moving averages

A simple²¹ moving average (MA) with length j at time t is calculated as the arithmetic mean of the j last prices up until time t . Mathematically we define it as follows:

$$MA_t(j) = \frac{1}{j} \sum_{i=0}^{j-1} P_{t-i} \quad (B.1)$$

In the overlapping MA trading strategy, a long (short) position is taken at time t if the short MA is higher (lower) or equal than the long MA at time t . This gives following trading signals for our MA strategies:

$$D_{MA,t+1} = \begin{cases} 1, & \text{for } MA_t(p) \geq MA_t(q) \\ -1, & \text{for } MA_t(p) < MA_t(q) \end{cases} \quad (B.2)$$

with $p < q$.

Appendix B.2. Relative strength indicator

The relative strength indicator (RSI) is a type of oscillator that takes values between 0 and 100. Traders use these values in order to determine oversold and overbought periods for a particular security. We calculate the RSI as follows:

$$RSI_t(h) = 100 - \frac{100}{1 + RS_t(h)} \quad (B.3)$$

with $RS = \frac{U_t(h)}{D_t(h)}$

Where $U_t(h)$ and $D_t(h)$ are the average number of periods with a positive or negative return in the last h periods, respectively. The RSI at time t indicates an overbought(oversold) period if it is v units higher(lower) than 50²². A buy(sell) signal is given when the oscillator moves out of the oversold(overbought) region. This results in following trade signals:

$$D_{RSI,t+1} = \begin{cases} 1, & \text{for } RSI_{t-1} \leq 50 - v \\ & \text{and } RSI_t > 50 - v \\ -1, & \text{for } RSI_{t-1} \geq 50 + v \\ & \text{and } RSI_t < 50 + v \\ 0, & \text{otherwise} \end{cases} \quad (B.4)$$

²⁰For a more extensive description we refer the reader to Sullivan et al. (1999) and Hsu et al. (2016) inter alia.

²¹Other types of MAs exist. Some of the more commonly used are: exponential (EMA), weighted (WMA) and filtered MAs.

²²Wilder (1978) suggests a lower bound value of 30 and an upper bound value of 70, corresponding with a v of 20. Usually, traders apply a v of 20 and an h of 14 days.

Appendix B.3. Support and resistance trading rules

Asset prices have certain levels, e.g. round numbers, around which orders tend to be clustered. As a result, crossing these values gives a strong indication of price momentum. Usually traders determine these levels by connecting past minima (support level) or past maxima (resistance level) graphically. As this is a quantitative study we cannot easily do qualitative analysis with graphs in order to determine these support and resistance levels. Consequently we use simple quantitative trading rule where we buy (sell) if the price moves above (below) a previous maximum (minimum).

Mathematically, if we define $\max_t(j)$ and $\min_t(j)$ as the maximum (resistance level) and minimum (support level) of the last j prices up until time t , then the trading signals given by the support and resistance trading rule are:

$$D_{\text{SR},t+1} = \begin{cases} 1, & \text{for } P_t > \max_{t-1}(j) \\ -1, & \text{for } P_t < \min_{t-1}(j) \\ 0, & \text{otherwise} \end{cases} \quad (\text{B.5})$$

Appendix B.4. Filter rules

Filter rules are another type of momentum indicator. A buy (sell) signal is given when the price moves $x\%$ ($y\%$) above its most recent low (high). Using $\max_t(j)$ and $\min_t(j)$, which we defined in Appendix B.3, we have:

$$D_{\text{F},t+1} = \begin{cases} 1, & \begin{aligned} &\text{for } P_t > (1+x)\min_{t-1}(j) \\ &\text{and } P_{t-1} \leq \min_{t-1}(j) \end{aligned} \\ -1, & \begin{aligned} &\text{for } P_t < (1-y)\max_{t-1}(j) \\ &\text{and } P_{t-1} \geq \max_{t-1}(j) \end{aligned} \\ 0, & \text{otherwise} \end{cases} \quad (\text{B.6})$$

We also use another type of filter rule where a buy (sell) signal is given if the price moves at least $x\%$ ($y\%$) above (below) its most recent low (high) and stays there for $d(x)$ ($d(y)$) days. The position is neutralized when the price moves $y\%$ ($x\%$) below (above) its subsequent most recent high (low) and stays there for $d(y)$ ($d(x)$) days. $y\%$ is less than $x\%$, $d(y)$ is less than or equal to $d(x)$.

Appendix B.5. Channel Breakouts

We define a $c\%$ -trading channel as a period where the most recent maximum price is within $c\%$ of the most recent minimum price. In other words, a $c\%$ -trading channel exists at time t if $(1-c)\max_t(j) \leq \min_t(j)$. When a trading channel exists, we apply support and resistance trading rules on the trading channel (see Appendix B.3 for the specific trading signals).

Appendix B.6. On-Balance Volume Averages

Research has shown that trading volume also contains valuable information. Technical traders pay close attention to trading volume in order to make trading decisions. Therefore, we find it prudent to also include a volume-based technical trading rule, namely the on-balance volume averages. We define the OBV_t , the on-balance volume at time t , as a running counter of the total volume traded up until time t . Namely:

$$\text{OBV}_t = \begin{cases} \text{OBV}_{t-1} + V_t, & \text{for } r_t \geq 0 \\ \text{OBV}_{t-1} - V_t, & \text{for } r_t < 0 \end{cases} \quad (\text{B.7})$$

Where V_t is the total volume traded between $t-1$ and t . In order to make trading decisions, we then apply moving average strategies to our OBV time series instead of our price time series, as described in Appendix B.1.

Appendix B.7. Parameters

TTR	Parameters	Description	Values
MA # 7,920	p	Length of short MA	1,2,6,12,18,24,30,48,96,144,168
	q	Length of long MA	2,6,12,18,24,30,48,96,144,168,192
	x	Fixed percentage band filter	0,0.05,0.1,0.5,1,5
	d	Time delay filter	0,2,3,4,5
	k	Holding period	6,12,24, ∞^a
RSI # 720	h	Lookback period of the RSI	2,6,12, 14,18,24,30,48,96,144,168,192
	v	$50 \pm v$ overbought/sold threshold	10,15,20,25
	d	Time delay filter	1,2,5
	k	Holding period	1,6,12,24, ∞^a
SR # 1,890	j	Lookback period	2,6,12,18,24,30,48,96,168
	x	Fixed percentage band filter	0.05,0.1,0.5,1,2.5,5,10
	d	Time delay filter	0,1,2,3,4,5
	k	Holding period	1,6,12,24, ∞^a
Filter # 1,575 + 1,260	j	Lookback period	1,2,6,12,24
	x	Filter for buy signals	0.05,0.1,0.5,1,5,10,20
	y	Filter for sell signals	0.05,0.1,0.5,1,5,10,20
	d(x)	Time delay filter for buy signals	0,1,2,3,4,5
	d(y)	Time delay filter for sell signals	0,1,2,3,4
	d	Uniform time delay filter	0,1,2,3,4,5
	k	Holding period	6,12,18,20,24, ∞^a
CB # 3,000	j	Lookback period	6,12,18,24,36,72,120,168
	c	c% trading channel	0.5,1,5,10, 15
	x	Fixed percentage band filter	0.05,0.1,0.5,1,5
	d	Time delay filter	0,1,2
	k	Holding period	1,6,12,24, ∞^a
OBV # 2,475	p	Length of short MA	2,6,12,18,24,30,48,96,144,168
	q	Length of long MA	2,6,12,18,24,30,48,96,144,168,192
	x	Fixed percentage band filter	0,0.01,0.05
	d	Time delay filter	0,2,3,4,5
	k	Holding period	6,12, ∞^a

Fixed percentage band filter: filter around the signal threshold in order to reduce noisy trades.
Time delay filter: Number of periods that a signal must be valid before a transaction is executed.
Holding period: Number of days a position is held, ignoring all other signals during that time.
 ∞^a meaning that the position will be held until a different trading signal is given.

Table B.8: Parameters used in the Technical Trading Rules.

The parameterization of our technical trading rules are based on earlier research by Sullivan et al. (1999) and Hsu et al. (2016). We largely use the same parameters as in Hsu et al. (2016), however we adjust the time lag parameters in order to make more sense at the 1 hour sample frequency (e.g. a MA length of 25 is changed to a length of 24). In addition to the parameters described earlier, we also apply 3 common parameters for every technical trading rule. In order to reduce the number of erroneous trades during volatile periods, we apply a fixed percentage band filter, x , around the trade signal thresholds, and a time delay filter, d , which represents the number of periods a trade signal should be valid before a trade is executed for every technical trading rule. In addition we also define a holding period, k , which represents the number of periods a trade position is held before returning to a neutral position. In total we compute 18,840 technical trading rules per sampling frequency, resulting in a total of 75,360 technical trading rules. Their individual parametrizations can be seen in table B.8.

Appendix C. FDR⁺ estimators

In this section we shortly replicate the methodology explained in Appendix B, C, D and E of Bajgrowicz and Scaillet (2012). In order to estimate the amount of positive false discoveries, FDR⁺, in our set of l

technical trading rules, we assume that the trading rules with no abnormal performance have uniformly distributed p-values over $[0,1]$. The FDR^+ is then defined as:

$$F\hat{D}R^+(\gamma) = \frac{\hat{F}^+}{\hat{R}^+} = \frac{\frac{1}{2}\hat{\pi}_0 l \gamma}{\#\{p_k \leq \gamma, \phi_k > 0; k = 1, \dots, l\}} \quad (C.1)$$

Where $\hat{\pi}_0$ is the estimated proportion of rules with no abnormal performance, γ ($\in [0,1]$) is a cut-off point for the p-values with abnormal performance, p_k and ϕ_k are the p-value and performance of trading rule k , respectively. $\hat{\pi}_0$ is estimated as follows:

$$\hat{\pi}_0(\lambda) = \frac{\#\{p_k > \lambda; k = 1, \dots, l\}}{l(1 - \lambda)} \quad (C.2)$$

Where $\lambda \in [0,1]$ is a tuning parameter which determines the minimum value after which all other are supposed to belong to non-abnormal trading rules. The selection of γ can be automated using the method of Storey (2002). In order to obtain a 10%- FDR^+ portfolio, we sort our trading rules with a positive performance from smallest to largest p-value. We add the rule with the smallest p-value and compute the FDR^+ if the 10% level is not obtained, we add the rule with the next highest p-value until we reach a FDR^+ of 10%.

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